

17:17

5G

37

<

Answer Details

1

2

3

4

5

Multiple Choice (Single Sel... 1 point

1

This test is totally the same as the mock exam sent via email and this test is provided for the convenience of viewing video explanations. Please note that the scoring rules for this version differ from the official exam and have no reference value. The test link sent via email is accurate. If you haven't completed the test via the email link yet, please do so first. Thank you!

A Yes I confirm.

Answer

A

Solution

Difficulty

★★★★★

think

17:18

5G 36



Answer Details

1

2

3

4

5

Multiple Choice (Single Sel... 1 point

0

What is the value of $(-2)^{-1} - (-2)^{-2} + (-2)^{-3}$?

A $-\frac{7}{8}$

B $-\frac{5}{8}$

C $-\frac{3}{8}$

D $\frac{3}{8}$

E $\frac{7}{8}$

Answer

A

Solution

Think Academy



17:18

5G 36



Answer Details

1

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5

Multiple Choice (Single Sel... 1 point

1



Town A and Town B are connected by a 240-mile road. A car leaves Town A and travels along this road toward Town B at a constant speed of 50 miles per hour. Sometime later, a second car leaves Town B and travels along the same road toward Town A at a constant speed of 60 miles per hour. The two cars pass each other exactly halfway between the two towns. If the first car left Town A at 1:00 PM, at what time did the second car leave Town B?

A 1:20 PM

B 1:24 PM

C 1:30 PM

D 1:40 PM

E 1:48 PM

Answer

B

17:18

5G 36



Answer Details

1

2

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5

Multiple Choice (Single Sel... 1 point

0



How many consecutive zeros are at the end of the decimal representation of the product $1! \cdot 2! \cdot 3! \cdots 25!$?

A 50

B 55

C 56

D 60

E 65

Answer

C

Solution

Think Academy



17:19

5G 35



Answer Details

1

2

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4

5

Multiple Choice (Single Sel... 1 point

1



The positive difference between the squares of two consecutive positive integers is a perfect cube greater than 1. What is the smallest possible value of the larger integer?

A 5

B 13

C 14

D 27

E 63

Answer

C

Solution

Think Academy

17:19

5G 35



Answer Details

6

7

8

9

10

Multiple Choice (Single Sel... 1 point

0



Let $N = \underbrace{111 \dots 1}_{2025 \text{ digits}}$. What is the sum of the digits of $2024 \times N$?

A 16192

B 16200

C 18216

D 18225

E 20250

Answer

B

Solution

Think Academy



17:19

5G 35



Answer Details

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Multiple Choice (Single Sel... 1 point

0

A large tank is initially full of water. Water continuously flows into the tank at a constant rate. When 3 identical pumps are turned on, they can empty the tank in 40 minutes. When 7 of these pumps are turned on, they can empty the tank in 10 minutes. If 5 of these pumps are turned on, how many minutes will it take to empty the tank?

A 14

B 16

C 20

D 24

E 25

Answer

B

Solution

17:19

5G 35



Answer Details

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Multiple Choice (Single Sel... 1 point

1



A subset of three distinct integers is chosen uniformly at random from the set $1, 2, 3, \dots, 10$. What is the probability that the smallest integer in the subset divides each of the other two, but neither of the larger two integers is a multiple of the other?

A $\frac{3}{40}$

B $\frac{7}{30}$

C $\frac{11}{40}$

D $\frac{17}{60}$

E $\frac{43}{120}$

Answer

D

Solution

17:20

5G 35



Answer Details

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Multiple Choice (Single Sel... 1 point

0

Positive integers x and y satisfy the equation $17x + 12y = 2024$. What is the greatest possible value of the greatest common divisor of x and y ?

A 22

B 44

C 46

D 88

E 92

Answer

B

Solution

Think Academy



17:20

5G 35



Answer Details

6

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Multiple Choice (Single Sel... 1 point

0



Real numbers x , y , and z satisfy $x \geq y \geq z$,
 $x + y + z = 3$, and $x^2 + y^2 + z^2 = 21$. What is
the maximum possible value of $x - z$?

A 5

B $3\sqrt{3}$

C 6

D $\sqrt{42}$ E $4\sqrt{3}$

Answer

C

Solution

Think Academy



20:44



Answer Details

11

12

13

14

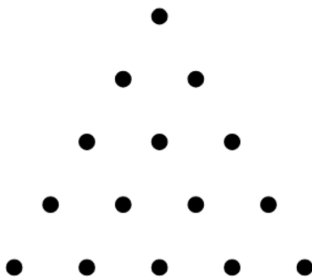
15

Multiple Choice (Single Sel... 1 point

0



Fifteen points are arranged in an equilateral triangular grid as shown, with one point in the first row, two points in the second row, and so on, up to five points in the fifth row. Three distinct points are chosen at random from the grid. What is the probability that the three points are collinear?



A $\frac{33}{455}$

B $\frac{9}{91}$

C $\frac{48}{455}$

D $\frac{51}{455}$

E $\frac{11}{91}$

20:44



Answer Details

11

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14

15

Multiple Choice (Single Sel... 1 point

0

List A consists of the 1000 integers of the form $5n + 2$ for $1 \leq n \leq 1000$. List B consists of the 1000 integers of the form $7n + 1$ for $1 \leq n \leq 1000$. The two lists are combined into a single list of 2000 integers, which is then sorted in nondecreasing order. What is the median of this combined list?

A 2919

B 2920

C 2921

D 2922

E 2923

Answer

C

Solution

11

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13

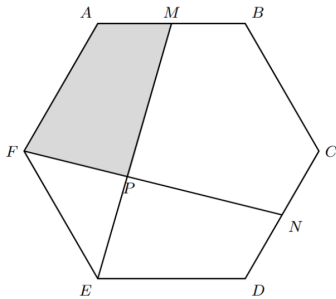
14

15

Multiple Choice (Single Sel... 1 point

0

Let $ABCDEF$ be a regular hexagon with side length 10. Let M be the midpoint of $\overline{AB}$, and let N be the midpoint of $\overline{CD}$. The line segments $\overline{EM}$ and $\overline{FN}$ intersect at P . What is the area of quadrilateral $AFPM$?



A $30\sqrt{3}$

B $\frac{125\sqrt{3}}{4}$

C $35\sqrt{3}$

D $\frac{75\sqrt{3}}{2}$

E $50\sqrt{3}$

20:44



Answer Details

11

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Multiple Choice (Single Sel... 1 point

0

A token starts at $(0, 0)$ on a coordinate plane. Each second, the token moves 1 unit in one of the four cardinal directions (north, south, east, or west), with each direction chosen independently and with equal probability. What is the probability that after 10 seconds, the token is at a point (x, y) such that $x - y$ is a multiple of 5?

A $\frac{125}{512}$

B $\frac{63}{256}$

C $\frac{127}{512}$

D $\frac{1}{4}$

E $\frac{129}{512}$

Answer

C



Answer Details

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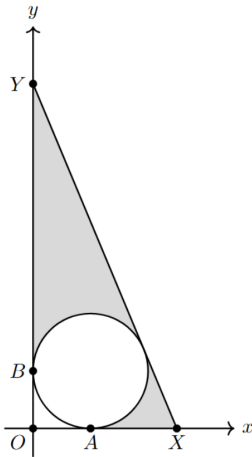
14

15

Multiple Choice (Single Sel... 1 point

0

A circle ω of radius 2 is tangent to the positive x -axis and positive y -axis at points A and B , respectively. A point X lies on the positive x -axis such that A is between the origin O and X , and $AX = 3$. The tangent to ω from X , other than the x -axis, intersects the y -axis at a point Y . What is the area of the region bounded by the segments AX , XY , YB , and the major arc AB of ω ?

A $26 - 4\pi$ B $26 - 3\pi$ C $30 - 4\pi$



Answer Details

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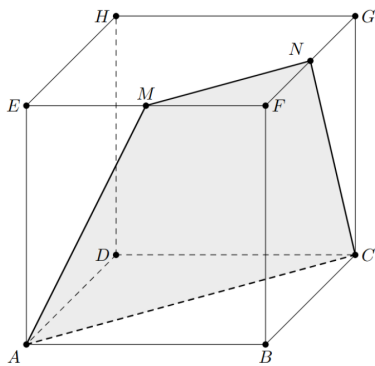
20

Multiple Choice (Single Sel... 1 point

0



Let $ABCD - EFGH$ be a cube with side length 6, where E, F, G , and H are the vertices directly above A, B, C , and D , respectively. Let M be the midpoint of $\overline{EF}$. A plane passing through A, C , and M divides the cube into two solids. What is the volume of the smaller solid?



A 36

B 54

C 63

D 72

E 108

20:45



Answer Details

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Multiple Choice (Single Sel... 1 point

0

How many positive integers $n \leq 200$ can be expressed as the difference of two positive perfect squares in exactly 3 distinct ways? (That is, there are exactly 3 pairs of positive integers (a, b) such that $a^2 - b^2 = n$.)

A 8

B 9

C 17

D 27

E 28

Answer

C

Solution

Let $n = a^2 - b^2 = (a - b)(a + b)$. Since a and b are positive integers, $a + b > a - b > 0$. Let $u = a - b$ and $v = a + b$. We require $0 < u < v$, $u \equiv v \pmod{2}$, and $uv = n$.

20:45



Answer Details

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Multiple Choice (Single Sel... 1 point

0



A sequence of four positive integers x_1, x_2, x_3, x_4 has the property that x_1, x_2, x_3 is an arithmetic progression and x_2, x_3, x_4 is a geometric progression. If $x_4 - x_1 = 80$, what is the sum of all possible values of x_1 ?

A 148

B 360

C 412

D 460

E 700

Answer

D

Solution

Let the geometric progression be $x_2 = kq^2$, $x_3 = kpq$, and $x_4 = kp^2$, where k, p , and q are positive integers and $\gcd(p, q) = 1$.

20:45



Answer Details

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Multiple Choice (Single Sel... 1 point

0



What is the remainder when $2^{2024} + 2^{2002} + 2025$ is divided by $2^{22} - 2^{11} + 1$?

A 2022

B 2023

C 2024

D 2025

E 2026

Answer

C

Solution

Let $x = 2^{11}$. The divisor can be rewritten as $x^2 - x + 1$.

We can also rewrite the dividend in terms of x .

Notice that $2^{2024} = (2^{11})^{184} = x^{184}$ and $2^{2002} = (2^{11})^{182} = x^{182}$.

Thus, we want to find the remainder when the

20:45



Answer Details

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Multiple Choice (Single Sel... 1 point

0



What is the sum of all positive integers b such that the base- b number 3025_b is a multiple of the base- b number 25_b ?

A 70

B 105

C 290

D 295

E 300

Answer

C

Solution

To find the valid bases b , we first convert the given numbers to base 10. The number 3025_b is equal to $3b^3 + 0b^2 + 2b + 5 = 3b^3 + 2b + 5$, and 25_b is equal to $2b + 5$. We are given that $2b + 5$ divides $3b^3 + 2b + 5$.



Answer Details

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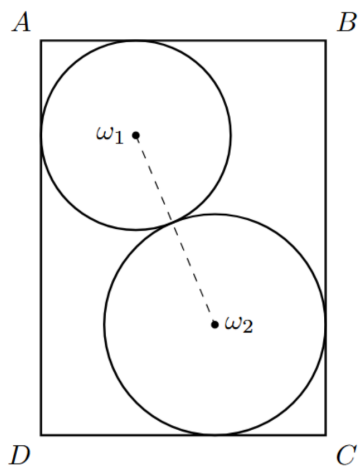
25

Multiple Choice (Single Sel... 1 point

1



Rectangle $ABCD$ has side lengths $AB = 18$ and $BC = 25$. Circle ω_1 is tangent to sides AB and AD , and circle ω_2 is tangent to sides CB and CD . The two circles lie entirely within the rectangle and are externally tangent to each other. Let M and m be the maximum and minimum possible values, respectively, of the sum of the areas of ω_1 and ω_2 . What is $M - m$?

A 12π B $\frac{25}{2}\pi$

20:45



Answer Details

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Multiple Choice (Single Sel... 1 point

0



Fourteen chairs are arranged in a row and numbered 1 through 14. The chairs are to be divided into 7 pairs so that the numbers of the two chairs in each pair differ by at most 3. In how many ways can this be done?

A 155

B 182

C 209

D 235

E 282

Answer

C

Solution

Let A_n be the number of valid pairings for $2n$ chairs. We want to find A_7 .

We classify the pairings of the set $1, 2, \dots, 2n$ by the chair paired with chair 1. Chair 1 can be

20:45



Answer Details

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Multiple Choice (Single Sel... 1 point

1



Let S be a subset of $\{1, 2, 3, \dots, 103\}$ such that for any two distinct elements $x, y \in S$, $|x - y|$ is neither 4 nor 7. What is the maximum possible number of elements in S ?

A 46

B 47

C 48

D 49

E 52

Answer

C

Solution

We want to find the maximum size of S such that no two elements differ by 4 or 7. Notice that $4 + 7 = 11$. This suggests analyzing the numbers modulo 11.

20:45



Answer Details

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Multiple Choice (Single Sel... 1 point

0

The numbers, in order from left to right, in each row of a 5×5 grid of positive integers form an arithmetic progression, and the numbers, in order from top to bottom, in each column form a geometric progression. Let $m_{i,j}$ denote the number in the i -th row and j -th column. If $m_{1,2} = 7$, $m_{2,3} = 20$, and $m_{3,4} = 52$, what is the value of $m_{5,2}$?

A 19

B 56

C 64

D 112

E 160

Answer

D

Solution

Let $m_{i,j}$ be the entry in row i and column j . Since the rows are arithmetic progressions and the

20:45



Answer Details

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Multiple Choice (Single Sel... 1 point

0

Let $f(0) = 0$, and for all integers $n \geq 0$,
 $f(2n) = 3f(n)$ and $f(2n + 1) = 3f(n) + 1$. Find
the number of integers n with $0 \leq n \leq 4095$ such
that $f(n)$ is a multiple of 7.

A 584

B 585

C 586

D 587

E 588

Answer

C

Solution

By computing the first few values of $f(n)$, we can
observe that $f(n)$ interprets the binary
representation of n as a number in base 3.

 k